



Currently.

A Secure Household Electricity Tracking Web App

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Overview

- Currently is a secure household electricity tracking web application designed to help users monitor, understand, and reduce their home electricity consumption.
- The system allows users to create rooms, assign appliances, view energy usage through interactive dashboards and generate estimated electricity costs and predictive insights based on their usage habits.

Motivation

Problem & Aim



Problem:

Many households struggle to understand which appliances and rooms drive their electricity usage and cost.



Aim:

Currently helps users track, visualise, and reduce household electricity consumption through a secure web application without requiring expensive extra hardware.

- Rising electricity costs and increasing household energy usage showed the need for a simpler and more user-friendly way for people to understand where electricity is being used and how they could potentially reduce their bills.
- I created the application to be accessible for the average homeowner, making sure there was no major technology barrier when learning about household energy usage and consumption.



How It Works

1



Sign Up / Log In

Create a secure account or log in to your existing account.

2



Add Floors & Rooms

Build your home digitally by adding floors and rooms that represent your space.

3



Assign Appliances

Choose appliances from our database and assign them to specific rooms.

4



Enter Usage Information

Input or estimate how long each appliance is used per day.

5



Calculate kWh & Cost

The system calculates your energy usage (kWh) and estimated electricity cost.

6



View Charts & Smart Insights

Explore visualisations, biggest energy users, forecasts, and personalised insights to save.



Tech Stack



Frontend



React



Vite



CSS



Vercel



Backend



Java



Spring Boot



Maven



REST APIs



Render



Database



PostgreSQL



JPA / Hibernate



Neon



Security



Spring Security



JWT



BCrypt



Tools



Postman



GitHub



Notion



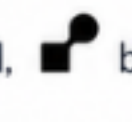
PlantUML



Deployment



Frontend deployed on Vercel,



backend deployed on Render,



database hosted on Neon.